Spectral Path Tracing in Shadertoy

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Abstract

We wrote a spectral path tracer in GLSL on the Shadertoy platform that implements dispersion, spectral upsampling, and thin film interference.

CCS Concepts: • **Computer Graphics → Light Transport.**

Keywords: Computer graphics, light transport, spectral rendering

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1. Introduction

Shadertoy (shadertoy.com) is an online platform and creative playground that allows users to write fragment shaders in GLSL that run on the GPU and generate visuals directly in the browser. We used Shadertoy to develop a spectral path tracer.

1.1. Motivation

A typical path tracer uses RGB triplets throughout rendering as a representation of color. This is an efficient approach as it uses the same representation that will be output to a display when a complete image is ready. It is, however, a simplification of the physical nature of light, which exhibits wave-like properties.

In the real world, color arises as a consequence of the varying wavelengths of light and how the human eye responds differently to different wavelengths. Before reaching an observer, light interacts with the environment, and many of these interactions are wavelength dependent, such as absorption which gives a surface color.

Some interactions like refraction and wave interference affect light in ways that are difficult to replicate in a path tracer which uses only RGB triplets to represent color. A spectral path tracer is designed to handle such phenomena by integrating in the spectral domain. It can also allow for more accurate color reproduction in certain scenarios.

2. Method

To produce a color we must integrate over the spectral domain to produce an RGB triplet. We do this by using the CIE 1931 XYZ color matching functions to first produce XYZ tristimulus values that we then convert to sRGB.

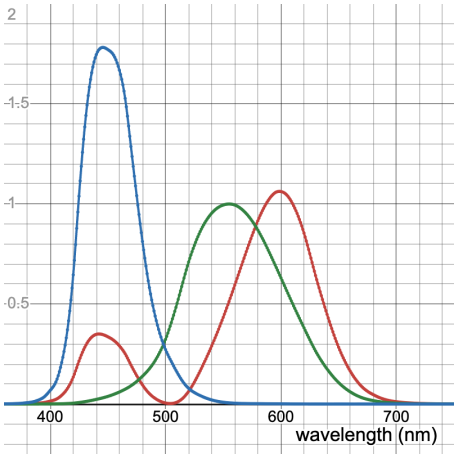


Figure 1: CIE 1931 XYZ color matching functions

$\overline{x}(\lambda)$, $\overline{y}(\lambda)$, and $\overline{y}(\lambda)$ are the color matching functions for a particular wavelength λ. To produce XYZ tristimulus values we use these and our rendering function $L(\boldsymbol{x}, \omega_o, \lambda)$ to integrate over the spectral domain:

$$X = \int_\lambda L(\boldsymbol{x}, \omega_o, \lambda)\overline{x}(\lambda)d\lambda$$

$$Y = \int_\lambda L(\boldsymbol{x}, \omega_o, \lambda)\overline{y}(\lambda)d\lambda$$

$$Z = \int_\lambda L(\boldsymbol{x}, \omega_o, \lambda)\overline{z}(\lambda)d\lambda$$

In practice, for each sample we randomly sample a wavelength from 360 to 740. At the end we determine the spectral radiance for that sample, which we multiple by the XYZ

tristimulus values for that wavelength and then add that to a total XYZ for the current pixel. The average XYZ tristimulus values for a pixel can be obtained by dividing this total by the number of samples rendered.

To finally produce an image in the sRGB colorspace, we operate on the average XYZ tristimulus values of the path traced output. The first step in this process is gamut clipping: for XYZ triplets that fall outside the sRGB color gamut, we follow a line of equal brightness to the D65 white point until we reach the edge of the color gamut. In practice we do not find many out of gamut colors in our renders, so the effect of this is rather minimal.

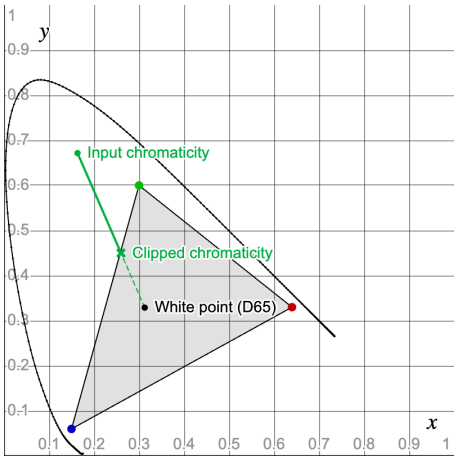


Figure 2: Gamut clipping to sRGB

Finally we perform a multiplication of the gamut clipped XYZ values arranged into a column vector by a 3x3 matrix to get a linear brightness sRGB triplet:

$$\begin{pmatrix} R \\ G \\ B \end{pmatrix} = \begin{pmatrix} 3.2406 & -1.5372 & -0.4986 \\ -0.9689 & 1.8758 & 0.0415 \\ 0.0557 & -0.2040 & 1.0570 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$$

With these values we can do final color adjustments before displaying to the screen.

2.1. Spectral Upsampling

We wanted to be able to define the colors of materials in the scene using RGB triplets. But in the spectral domain, colors are defined with curves that vary over wavelength. So to use RGB triplets in a spectral renderer, we must reconstruct a curve whose color is the same or as close as possible to the color represented by the RGB triplet.

To define these curves we use the function outlined in Jakob [2], which combines a parabola and a sigmoid:

$$S(x) = \frac{1}{2} + \frac{x}{2\sqrt{1+x^2}}$$

$$S(c_0 x^2 + c_1 x + c_2)$$

The parameters c_1, c_2, c_3 must be optimized to match an RGB color of interest. Jakob and Hanika used an offline solver to generate lookup tables to map RGB values to coefficients for a number of colorspaces.

To find coefficients matching RGB values chosen for materials in a previous version of the renderer that used RGB, we used their code to generate them, and in the spectral path tracer accepted those values instead of RGB.

Another consideration is that this function space is designed to fit the constraint of a reflectance curve, which is that at no point should it exceed one, respecting conservation of energy. To use these curves for emissive materials, which usually need values higher than one, we multiply the outputs sampled from these functions by an intensity value which can be tuned per material.

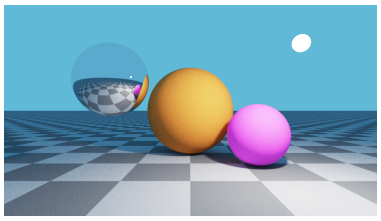


Figure 3: Previous result from RGB path tracer.

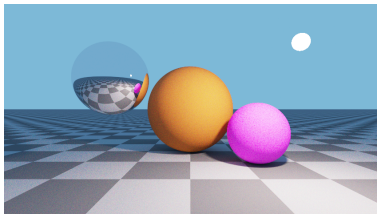


Figure 4: New result with spectral upsampling.

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2.2. Dispersion

Dispersion is a phenomena where light will refract at differing angles through an interface, relating to how refractive indices can vary by wavelength. In our renderer we use the two-term form of Cauchy's equation for refractive index as a simple model of this behavior. Here λ is the wavelength and A, B are specialized coefficients:

$$n(\lambda) = A + \frac{B}{\lambda^2}$$

2.3. Thin Film Interference

For our scene we wanted to render a soap bubble. Soap bubbles exhibit color variations that are the result of thin film interference. Thin film interference is an optical phenomenon that occurs when light reflects from the top and bottom surfaces of a very thin layer of material, such as oil on water or a soap bubble. Because the film's thickness is comparable to the wavelength of visible light, the reflected waves can interfere constructively or destructively depending on their relative phase. This phase difference arises from both the path length traveled within the film and any phase shifts that occur upon reflection at boundaries with different refractive indices. Because of this, certain wavelengths are enhanced while others are suppressed, producing vivid color patterns.

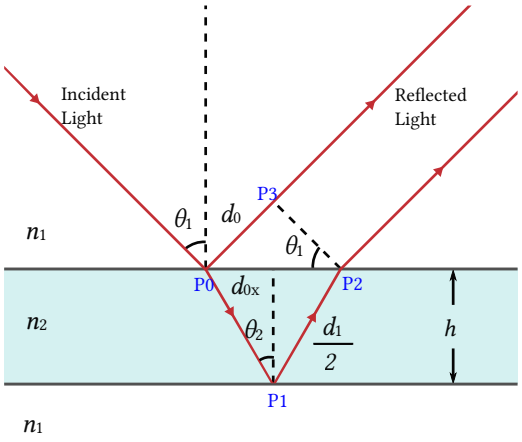


Figure 5: Diagram of thin film interference. (Modified from [1])

Given this diagram where h is the depth of the thin film, θ_2 is the angle at which the ray refracts, and λ_2 is the wavelength of the ray. The distance traveled by the reflected ray is d_0 and the distance traveled by the refracted ray is d_1. A formula from these distances can be used to determine the phase shift of the wave.

2.3.1. Derivation of Phase Shift

The majority of the missing values from the diagram can be obtained by solving a triangulation problem where its known what s_x, θ, O (origin, bottom left), p_x (bottom right) and p (top right) are, but s_y and s are unknown. This outlines solving for these values for use later on. Note that for these derivations we try to express everything in $\cos(\theta)$ as that is the dot product of the ray and the normal, which decreases the amount of computation.

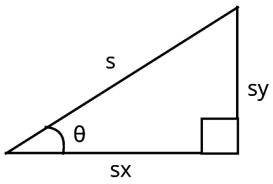


Figure 6: Diagram of triangulation problem.

$$p_x = (s_x, 0)$$

$$p = (\cos(\theta), \sin(\theta)) * \frac{s_x}{\cos(\theta)}$$

$$p = \left(s_x, s_x * \frac{\sin(\theta)}{\cos(\theta)}\right)$$

Find s_y

$$s_y(s_x, \theta) = s_x * \frac{\sin(\theta)}{\cos(\theta)}$$

$$s_y(s_x, \theta) = s_x * \frac{\sqrt{1-\cos^2(\theta)}}{\cos(\theta)}$$

Find s

$$s(s_x, \theta) = \sqrt{s_x^2 + \left(\frac{s_x \sin(\theta)}{\cos(\theta)}\right)^2}$$

$$s(s_x, \theta) = \sqrt{s_x^2 + \frac{s_x^2 \sin^2(\theta)}{\cos^2(\theta)}}$$

$$s(s_x, \theta) = \sqrt{s_x^2\left(1 + \frac{\sin^2(\theta)}{\cos^2(\theta)}\right)}$$

$$s(s_x, \theta) = s_x \sqrt{1 + \frac{\sin^2(\theta)}{\cos^2(\theta)}}$$

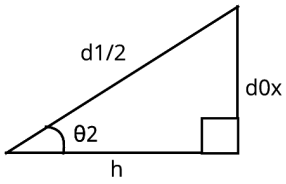
$$s(s_x, \theta) = s_x \sqrt{1 + \tan^2(\theta)}$$

$$s(s_x, \theta) = s_x \sqrt{\sec^2(\theta)}$$

$$s(s_x, \theta) = s_x \sec(\theta)$$

$$s(s_x, \theta) = \frac{s_x}{\cos(\theta)}$$

Now that we have defined this problem and abstracted it we can now use them to find missing values in the previous diagram shown of thin film interference. Thus we apply the functions created.



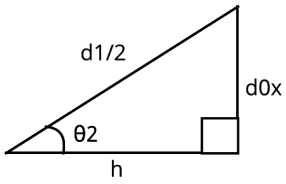
$$\frac{d_1}{2} = s(h, \theta_2)$$

$$\frac{d_1}{2} = \frac{h}{\cos(\theta_2)}$$

$$d_1 = \frac{2h}{\cos(\theta_2)}$$

$$d_{0_x} = s_{y(h,\theta_2)}$$

$$d_{0_x} = h * \frac{\sqrt{1-\cos^2(\theta_2)}}{\cos(\theta_2)}$$



$$d_0 = s\left(d_{0_x}, \theta_1\right)$$

$$d_0 = \frac{d_{0_x}}{\cos(\theta_1)}$$

$$d_0 = h * \frac{\sqrt{1-\cos^2(\theta_2)}}{\cos(\theta_2)\cos(\theta_1)}$$

Now we can compute phase shift b. Let ior be denoted by n.

$$b = (d_1 n - d_0) * \frac{2\pi}{\lambda}$$

$$b = \left(\frac{2hn}{\cos(\theta_2)} - \frac{h\sqrt{1-\cos^2(\theta_2)}}{\cos(\theta_2)\cos(\theta_1)}\right) * \frac{2\pi}{\lambda}$$

$$b = \left(\frac{h\pi}{\cos(\theta_2)\lambda}\right)\left(2n - \frac{\sqrt{1-\cos^2(\theta_2)}}{\cos(\theta_1)}\right)$$

2.3.2. Derivation of Intensity of Light

Now we can find the outgoing intensity. Let a be the intensity of the outgoing wave. Let I_i be the intensity of the ingoing wave. The outgoing wave is comprised of two waves, the reflected wave from the ingoing wave, and the refracted wave from the ingoing wave. Let I_0 be the intensity of the reflected wave and let I_1 be the intensity of the refracted wave. Let f_x be defined as the fresnel function for the intersection at point Px where there is a change in the index of refraction of two separate materials. We can get the intensity of the waves through multiplication of fresnel constants.

$$a_0 = f_0$$
$$a_1 = (1 - f_0) f_1 (1 - f_2)$$
$$I_0 = a_0 I_i$$
$$I_1 = a_1 I_i$$

We can now define the following the incoming wave as $I_i \sin(x)$. The reflected component of the outgoing wave as $I_0 \sin(x)$. The refracted then reflected component of the outgoing wave as $I_1 \sin(x + b)$ where b is the phase shift. And the outgoing wave as $I_0 \sin(x) + I_1 \sin(x + b) = a \sin(x + c)$. The outgoing wave formula is where this de-

rivation will start. This derivation details finding a which is the outgoing intensity.

Step 1:

$I_0 \sin(x) + I_1 \sin(x+b) = a \sin(x+c)$

Step 2. Apply $\sin(l+k) = \sin l \cos k + \cos k \sin l$

$I_0 \sin(x) + I_1 \sin(x)\cos(b) + I_1 \cos(x)\sin(b) = a \sin(x+c)$

Step 3. Apply $\sin(l+k) = \sin l \cos k + \cos k \sin l$

$I_0 \sin(x) + I_1 \sin(x)\cos(b) + I_1 \cos(x)\sin(b) = a \sin(x)\cos(c) + a\cos(x)\sin(c)$

Step 4. Match Coefficients From Step 3

$(I_0 + I_1 \cos(b))\sin(x) + (I_1 \sin(b))\cos(x) = (a\cos(c))\sin(x) + (a\sin(c))\cos(x)$

Step 5 Match Coefficients for sin(x) From Step 3.

$I_0 + I_1 \cos(b) = a\cos(c)$

Step 6. Match Coefficients for cos(x) From Step 3.

$I_1 \sin(b) = a \sin(c)$

Step 7. From 6

$\frac{I_1 \sin(b)}{a} = \sin(c)$

Step 8. From 7

$\therefore c = \arcsin\left(\frac{I_1 \sin(b)}{a}\right)$

Step 9. Apply Power of 2 to Step 5

$(I_0 + I_1 \cos(b))^2 = a^2 \cos^2(c)$

Step 10. Apply Trig Identity

$(I_0 + I_1 \cos(b))^2 = a^2(1 - \sin^2(c))$

Step 11. Substitute sin(c) From 7 Into 10

$(I_0 + I_1 \cos(b))^2 = a^2\left(1 - \left(\frac{I_1 \sin(b)}{a}\right)^2\right)$

Step 12. Simplify

$(I_0 + I_1 \cos(b))^2 = a^2 - I_1^2 \sin^2(b)$

Step 13. Add $I_1^2 \sin^2(b)$

$(I_0 + I_1 \cos(b))^2 + I_1^2 \sin^2(b) = a^2$

Step 14. Distribute

$I_0^2 + 2I_0 I_1 \cos(b) + I_1^2 \cos^2(b) + I_1^2 \sin^2(b) = a^2$

Step 15. Apply $\cos^2(x) + \sin^2(x) = 1$

$I_0^2 + 2I_0 I_1 \cos(b) + I_1^2 = a^2$

Step 16. Substitute I_0 and I_1

$(a_0 I_i)^2 + 2(a_0)(a_1 I_i)\cos(b) + (a_1 I_i)^2 = a^2$

Step 17. Factor I_i^2

$I_i^2(a_0^2 + 2a_0 a_1 \cos(b) + a_1^2) = a^2$

Step 18. Sqrt

$\therefore a = I_i\sqrt{a_0^2 + 2a_0 a_1 \cos(b) + a_1^2}$

Now this intensity can be applied to each ray that hits the thin film material.

3. Results

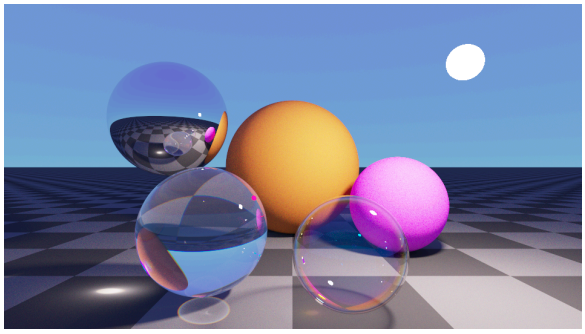


Figure 9: Render at 940x540

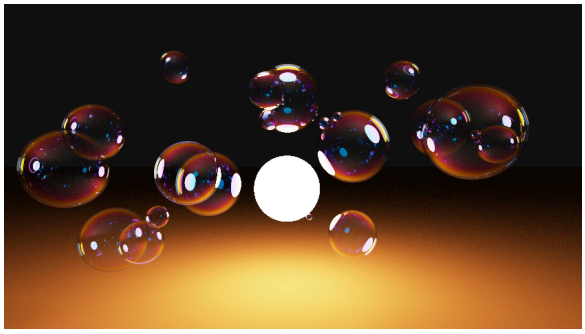


Figure 10: Render of many bubbles at 840x472

You can also render this scene live in the browser and preview all the shader code used with this link: https://www.shadertoy.com/view/WfdBDH. The majority of the code is in Buffer A (buffer_a.frag.glsl)

3.1. Division of Work

Julian worked on dispersion, spectral upsampling and the rendering engine for the shader. Samuel worked on derivation of formulas for thin film interference and its implementation into the path tracing engine.

References

[1] Wikimedia Commons. 2021. Thin film interference.svg. Retrieved from https://commons.wikimedia.org/w/index.php?title=File:Thin_film_interference.svg&oldid=548357603

[2] Wenzel Jakob and Johannes Hanika. 2019. A Low-Dimensional Function Space for Efficient Spectral Upsampling. *Computer Graphics Forum (Proceedings of Eurographics)* 38, 2 (March 2019).